Project: Worked examples
Title: O-ring and seal gland
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1. Scope and method

Squeeze, stretch, gland fill and extrusion gap of a radial piston, radial rod or face O-ring seal, from the ring and groove dimensions with their tolerances, at the cold and shrunk and at the hot and swollen end of the temperature and swell range. The ring's cross-section shrinks with stretch by a relation you choose (none, constant-volume rubber, the quadratic cited from Parker ORD 5700 section 3.5, or linear in stretch). The worst case and the capability indices run through the tolerance-stack engine of M1, and the seeded Monte Carlo of the engine runs the exact non-linear model and reports the fraction outside each limit. A face seal is entered with the pressure on its outside or its inside. Measured groove distributions (from the scan import) can replace the toleranced dimensions. Ring sizes, the squeeze and fill bands and the extrusion limit are your own records: no O-ring size table, gland table or extrusion chart is shipped.
Stretch: $X = (g \cdot D_{ref})/(f \cdot d) - 1$, D_{ref} the diameter the ring ID sits on (groove root for a piston or face seal, rod for a rod seal); $f = (1 + \alpha(T - 20\ °C))(1 + swell)^{(1/3)}$ the ring and $g = 1 + \alpha_m(T - 20\ °C)$ the groove metal Cross-section: $W_{eff} = f \cdot W \cdot (1 - r(X))/\sqrt{(1 - c)}$, $r(X)$ the chosen relation (none; $1 - 1/\sqrt{(1+X)}$; $Y = 0.56 + 0.59X - 0.0046X^2$ per cent as cited, used for about 3 to 25 % stretch only; $v \cdot X$) and c the compression of the ring outside diameter where it is held (rod seal in the housing groove; internal-pressure face seal on the groove outer wall), $c = 1 - g \cdot D_g/(f \cdot (d + 2W))$ Depth and gap: piston $h = (D_s - D_g)/2$, gap $= D_s - D_c$; rod $h = (D_g - D_s)/2$, gap $= D_c - D_s$; face h and gap entered directly (all times g) Squeeze: $q = 1 - g \cdot h/W_{eff}$, minimum at the cold end from the largest depth over the smallest W_{eff}, maximum at the hot end Gland fill: ring area over the groove area $g \cdot b \cdot g \cdot h - 2(1 - \pi/4)(g \cdot r_g)^2$; maximum at the hot end on a volume basis, $(\pi/4) \cdot (f \cdot W)^2 \cdot (f/g)$ with $f^3 = (1 + \alpha\Delta T)^3(1 + s)$ (an outside diameter held in the groove: $(\pi/4) \cdot (f \cdot W)^2/(1 - c)$), minimum at the cold end $(\pi/4) \cdot W_{eff}^2$ Worst case: each range from the engine's worst-case stack of the dimension ($\Sigma|coefficient| \cdot$half-band), combined at the corner that is worst for the quantity Monte Carlo: the engine's seeded Monte Carlo on the exact criterion, each dimension from its own distribution; the fraction below the lower and above the upper limit. The capability index C_{pk} uses the criterion linearised about the mean dimensions (secant slope across each band) Extrusion: the largest gap over the installed, cold and hot temperatures (the sealing-surface part and the groove metal each with its own expansion) plus the bore dilation under pressure, against the permitted gap from your record (interpolated in hardness and pressure) or the value entered; with neither the check fails (not verified)
Ref: Parker O-Ring Handbook ORD 5700 Section 3.5 — Cross-section reduction with stretch (relation quoted in Addendum B; not confirmed)
Ref: Parker PTD 5705 — O-ring installation guidance (stretch rule quoted in Addendum B; not confirmed)
Ref: ISO 3601-1 — O-rings: inside diameters, cross-sections, tolerances and size codes
Ref: ISO 3601-2 — O-rings: housing dimensions for general applications

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Gland arrangement	Piston (radial)		spec: Cylinder datasheet LC-63: ring on the piston, sealing on the bore	any
duty	Static or dynamic duty (selects the squeeze band)	Static		spec: Cylinder datasheet LC-63: locking duty, no stroke under pressure	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
face_p	Face seal only: which side the pressure acts on (external: the ring ID seats on the groove inner wall and is stretched; internal: the ring OD seats on the groove outer wall and is compressed)	External (ring ID seated on the groove)		default	any
d_r	Ring inside diameter d (mean of the supplier's tolerance)	56	mm	reference: record: oring.ring, illustrative ring size invented for this example, to be replaced by the supplier's catalogue size	3 … 500 mm
d_r_tol	Ring inside diameter tolerance, plus or minus	0.45	mm	reference: record: oring.ring, illustrative tolerance invented for this example	0 … 2 mm
W	Ring cross-section diameter W	3.5	mm	reference: record: oring.ring, illustrative cross-section invented for this example	1 … 10 mm
W_tol	Ring cross-section tolerance, plus or minus	0.1	mm	reference: record: oring.ring, illustrative tolerance invented for this example	0 … 0.4 mm
D_s	Sealing-surface diameter: the bore of a piston seal, the rod of a rod seal (not used for a face seal)	63	mm	drawing: LC-63-001 rev B	5 … 1000 mm
D_s_u	Sealing-surface diameter, upper deviation	0.046	mm	drawing: LC-63-001 rev B	-0.5 … 0.5 mm
D_s_l	Sealing-surface diameter, lower deviation	0	mm	drawing: LC-63-001 rev B	-0.5 … 0.5 mm
D_g	Groove root diameter: on the piston, in the housing of a rod seal, or for a face seal the diameter the ring seats on (the groove inner wall for external pressure, the groove outer wall for internal pressure)	57.5	mm	drawing: LC-63-002 rev B	5 … 1000 mm
D_g_u	Groove root diameter, upper deviation	0	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
D_g_l	Groove root diameter, lower deviation	-0.074	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
D_c	Mating part that sets the extrusion gap: the piston outside diameter, or the housing bore round the rod (not used for a face seal)	63	mm	drawing: LC-63-002 rev B	5 … 1000 mm
D_c_u	Mating part diameter, upper deviation	-0.03	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
D_c_l	Mating part diameter, lower deviation	-0.06	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
h_f	Face-seal groove depth (face seal only)	2.7	mm	default	0.3 … 20 mm
h_f_u	Face-seal groove depth, upper deviation	0.05	mm	default	-0.5 … 0.5 mm
h_f_l	Face-seal groove depth, lower deviation	0	mm	default	-0.5 … 0.5 mm
e_f	Face-seal extrusion gap between the flanges (face seal only)	0.05	mm	default	0 … 2 mm
e_f_u	Face-seal gap, upper deviation	0.05	mm	default	-0.5 … 0.5 mm
e_f_l	Face-seal gap, lower deviation	0	mm	default	-0.5 … 0.5 mm
b	Groove width b	4.7	mm	drawing: LC-63-002 rev B	1 … 20 mm
b_u	Groove width, upper deviation	0.1	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
b_l	Groove width, lower deviation	0	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
r_g	Groove bottom corner radius	0.25	mm	drawing: LC-63-002 rev B	0 … 1 mm
T_min	Lowest operating temperature (the ring sits at 20 °C when installed)	-25	degC	spec: Hydraulic specification HS-63: oil temperature range −25 to +70 °C	-60 … 100 degC
T_max	Highest operating temperature	70	degC	spec: Hydraulic specification HS-63: oil temperature range −25 to +70 °C	-40 … 200 degC

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
alpha	Ring linear thermal expansion coefficient α (an indicative elastomer figure; take yours from the compound or the material library)	0.00015	1/K	assumption: Indicative elastomer figure; replace by the compound's value or a material from the library.	0.00005 … 0.0003 1/K
alpha_m	Groove metal linear thermal expansion coefficient α_m (the part that carries the groove; one material record per sheet, so this one is entered from the metal's data sheet)	0.000012	1/K	assumption: Steel groove metal, indicative.	0.000005 … 0.00003 1/K
s_min	Smallest volume change of the ring in the fluid (negative for shrinkage; 0 if none)	0	%	assumption: No shrinkage assumed in the mineral oil.	-10 … 0 %
s_max	Largest volume swell of the ring in the fluid	4	%	assumption: Volume swell of the compound in the oil, assumed 4 %; replace by the compound's immersion data.	0 … 25 %
red	Cross-section reduction with stretch	Incompressible rubber (derived)		assumption: Constant-volume rubber relation, derived.	any
nu	Poisson ratio of the ring, for the linear relation	0.49		default	0.3 … 0.5
q_lo_s	Static duty: lowest acceptable squeeze (from your handbook)	15	%	reference: record: oring.criteria, static squeeze band, to be taken from the owner's handbook or the seal maker	5 … 30 %
q_hi_s	Static duty: highest acceptable squeeze	30	%	reference: record: oring.criteria, static squeeze band, to be taken from the owner's handbook or the seal maker	10 … 45 %
q_lo_d	Dynamic duty: lowest acceptable squeeze	10	%	default	5 … 25 %
q_hi_d	Dynamic duty: highest acceptable squeeze	25	%	default	10 … 35 %

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
fill_lo	Gland fill: lowest acceptable	60	%	reference: record: oring.criteria, gland fill band (Addendum B default 60 to 85 %)	40 … 80 %
fill_hi	Gland fill: highest acceptable	85	%	reference: record: oring.criteria, gland fill band (Addendum B default 60 to 85 %)	60 … 95 %
x_lo	Installed stretch: lowest acceptable (negative allows a ring that is compressed on its seat)	0	%	reference: record: oring.criteria, installed stretch lower limit	-5 … 5 %
x_hi	Installed stretch: highest acceptable (the installation rule is up to 6 % on a piston groove); also applied at the cold end	6	%	reference: record: oring.criteria, installed stretch limit (Addendum B quotes up to 6 % on a piston groove)	1 … 10 %
od_hi	Ring outside-diameter compression: highest acceptable, in a rod-seal groove or against the groove outer wall of an internal-pressure face seal (your record; the default is a placeholder equal to the stretch limit, not a handbook value)	6	%	default	1 … 10 %
H_A	Ring hardness, Shore A (for the extrusion record)	70		spec: Seal specification: compound 70 Shore A	40 … 95
p_op	Operating pressure (for the extrusion record)	160	bar	spec: Hydraulic specification HS-63: working pressure 160 bar	0 … 700 bar
dD_p	Diametral growth under the operating pressure of the bore (piston seal) or of the housing bore round the rod (rod seal), from your cylinder or housing calculation; added to the extrusion gap (0 if not known)	0	mm	default	0 … 0.5 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
gap_allow	Largest permitted extrusion gap at this pressure and hardness, if you have it as one value	0.2	mm	reference: record: oring.extrusion, illustrative permitted gap invented for this example, to be replaced by the seal maker's value for 70 Shore A at 160 bar	0.01 … 1 mm
n_mc	Monte Carlo samples	20000		assumption: 20 000 samples.	1000 … 100000
seed	Monte Carlo seed	42		assumption: Fixed seed for a repeatable sheet.	0 … 4295000000
k_sigma	Standard deviations per half-band (normal tolerances and measured normals)	3		assumption: Normal tolerances taken as ±3σ.	1 … 6
dist	Distribution of the toleranced dimensions	Normal		assumption: Normal distribution of the machined dimensions.	any

3. Assumptions

A1. The gland is a plain rectangular groove with a bottom corner radius; the ring is a circle of the stated cross-section. Edge breaks, groove wall angles and surface finish are not modelled. (Closed-form gland geometry; no gland table.)
A2. Squeeze is the compressed fraction of the effective cross-section W_eff, which is the nominal section after thermal expansion, swell and the cross-section reduction of the stretch. The reduction relation is chosen by the user; the quadratic of Parker ORD 5700 section 3.5 is implemented as quoted in Addendum B, used for about 3 to 25 % stretch only, and is not confirmed against the handbook. Where the ring outside diameter is held (a rod seal in its housing groove, an internal-pressure face seal), the compression c thickens the section to W/√(1 − c). (The relation is an input choice; the default is the constant-volume rubber relation, derived.)
A3. The cold end is the lowest temperature with the least swell and sets the minimum squeeze and fill; the hot end is the highest temperature with the most swell and sets the maximum. This assumes the ring expands more than the groove metal (alpha above alpha_m). (Bounding cases of a monotonic model.)
A4. The worst case combines each dimension at its extreme independently. Where one dimension appears twice (the groove root diameter sets both the depth and the stretch of a piston seal) the worst case is slightly conservative. (Independent ends of linear stacks.)
A5. The Monte Carlo samples each dimension from its own distribution about its mean and evaluates each criterion exactly (the non-linear model) on the engine's seeded sampler. It counts the fraction outside each limit at the cold or hot end; it does not model pressure, friction or compression set. The capability index C_pk uses the first-order (secant) linearisation. (Exact model on the engine's seeded sampler.)
A6. The gland fill at the hot end is on a volume basis: a ring held circumferentially by the groove takes the whole volume change of thermal expansion and swell in its section, $(1 + \alpha\Delta T)^3(1 + s)$, while the groove circumference grows with the groove metal. This is higher than scaling the section area isotropically. (Volume conservation of the held ring.)
A7. The extrusion gap is the largest diametral clearance of a radial seal (the axial gap of a face seal) over the installed, cold and hot temperatures, the sealing-surface part and the groove metal each with its own

expansion coefficient, plus the diametral growth of the bore under pressure that the user enters (0 if none). The permitted gap is the user's record; the sheet never states a handbook value. With no permitted gap entered the extrusion check fails (not verified) while the operating pressure is above zero. (Extrusion limits are licensed data and are not shipped.)
A8. alpha (Ring linear thermal expansion coefficient α (an indicative elastomer figure; take yours from the compound or the material library)) is taken as 0.00015 1/K. (Indicative elastomer figure; replace by the compound's value or a material from the library.)
A9. alpha_m (Groove metal linear thermal expansion coefficient α_m (the part that carries the groove; one material record per sheet, so this one is entered from the metal's data sheet)) is taken as 0.000012 1/K. (Steel groove metal, indicative.)
A10. s_min (Smallest volume change of the ring in the fluid (negative for shrinkage; 0 if none)) is taken as 0 %. (No shrinkage assumed in the mineral oil.)
A11. s_max (Largest volume swell of the ring in the fluid) is taken as 4 %. (Volume swell of the compound in the oil, assumed 4 %; replace by the compound's immersion data.)
A12. red (Cross-section reduction with stretch) is taken as Incompressible rubber (derived). (Constant-volume rubber relation, derived.)
A13. n_mc (Monte Carlo samples) is taken as 20000. (20 000 samples.)
A14. seed (Monte Carlo seed) is taken as 42. (Fixed seed for a repeatable sheet.)
A15. k_sigma (Standard deviations per half-band (normal tolerances and measured normals)) is taken as 3. (Normal tolerances taken as ±3σ.)
A16. dist (Distribution of the toleranced dimensions) is taken as Normal. (Normal distribution of the machined dimensions.)
A17. Static duty: the squeeze band for static seals from the oring.criteria record is used. (The bands are the user's record.)

4. Working

[41] v_W = mid ± half-band, □ (normal)
= 3.5 mm nominal → mid 3.5 ± 0.1 mm, □ 0.033333 mm
= 3.5 mm
ring cross-section W

[42] v_d = mid ± half-band, □ (normal)
= 56 mm nominal → mid 56 ± 0.45 mm, □ 0.15 mm
= 56 mm
ring inside diameter d

[43] v_Ds = mid ± half-band, □ (normal)
= 63 mm nominal → mid 63.023 ± 0.023 mm, □ 0.0076667 mm
= 63.023 mm
bore D_s

[44] v_Dg = mid ± half-band, □ (normal)
= 57.5 mm nominal → mid 57.463 ± 0.037 mm, □ 0.012333 mm
= 57.463 mm
piston groove root D_g

[45] v_Dc = mid ± half-band, □ (normal)
= 63 mm nominal → mid 62.955 ± 0.015 mm, □ 0.005 mm
= 62.955 mm
piston outside diameter D_c

[46] v_b = mid ± half-band, □ (normal)
= 4.7 mm nominal → mid 4.75 ± 0.05 mm, □ 0.016667 mm
= 4.75 mm
groove width b

[47] f_c = (1 + □·(T_min − 20 °C))·(1 + s_min)^(1/3)
= (1 + 0.00015·(-25 − 20))·(1 + 0)^(1/3)
= 0.99325

Ring size factor, cold end

[48] g_c = 1 + □_m·(T_min − 20 °C)
= 1 + 0.000012·(-25 − 20)
= 0.99946

Groove metal size factor, cold end

[49] f_h = (1 + □·(T_max − 20 °C))·(1 + s_max)^(1/3)
= (1 + 0.00015·(70 − 20))·(1 + 0.04)^(1/3)
= 1.0208

Ring size factor, hot end

[50] g_h = 1 + □_m·(T_max − 20 °C)
= 1 + 0.000012·(70 − 20)
= 1.0006

Groove metal size factor, hot end

[51] stretch_nom = D_ref/d − 1 at the mean dimensions
= 57.463/56 − 1
= 2.6125 %

Installed stretch of the ring ID on the seat, mean dimensions

[52] stretch = D_ref,max/d_min − 1
= 57.5/55.55 − 1
= 3.5104 %

Largest installed stretch

[53] stretch_min = D_ref,min/d_max − 1
= 57.426/56.45 − 1
= 1.729 %

Smallest installed stretch

[54] red_nom = r(X) at the mean installed stretch
= r(2.6125 %)
= 1.2812 %

Cross-section reduction at the mean installed stretch

[55] u_stretch = stretch / x_hi
= 3.5104 % / 6 %
= 0.58506

Largest installed stretch against its limit

[56] X_c = g_c·D_ref,max/(f_c·d_min) − 1
= 0.99946·57.5/(0.99325·55.55) − 1
= 4.1575 %

Largest stretch at the cold end

[57] r_c = r(X_c)
= r(4.1575 %)
= 2.0161 %

Cross-section reduction at the cold end

[58] Wc_lo = f_c·W_min
= 0.99325·3.4
= 3.377 mm

Smallest ring section at the cold end

[59] Weff_lo = Wc_lo·(1 − r_c)
= 3.377·(1 − 0.020161)
= 3.309 mm

Smallest effective ring section

[60] h_hi_c = g_c·(depth stack) maximum
= mean 2.7785 mm, worst-case maximum
= 2.8085 mm

Largest gland depth at the cold end

[61] squeeze_min = 1 - h_hi_c/Weff_lo
= 1 - 2.8085/3.309
= 15.125 %
Smallest squeeze
[62] A_g_hi = b_max·h_max - 2(1 - π/4)·(g_c·r_g)²
= 4.7974·2.8085 - 0.026796
= 13.447 mm²
Largest groove area at the cold end
[63] fill_min = (π/4)·Weff_lo²/A_g_hi
= (π/4)·3.309²/13.447
= 63.953 %
Smallest gland fill
[64] X_h = g_h·D_ref,min/(f_h·d_max) - 1
= 1.0006·57.426/(1.0208·56.45) - 1
= -0.28 %
Smallest stretch at the hot end
[65] r_h = r(X_h)
= r(-0.28 %)
= -0.14029 %
Cross-section reduction at the hot end
[66] Wh_hi = f_h·W_max
= 1.0208·3.6
= 3.6747 mm
Largest ring section at the hot end
[67] Weff_hi = Wh_hi·(1 - r_h)
= 3.6747·(1 - -0.0014029)
= 3.6799 mm
Largest effective ring section
[68] h_lo_h = g_h·(depth stack) minimum
= mean 2.7817 mm, worst-case minimum
= 2.7516 mm
Smallest gland depth at the hot end
[69] squeeze_max = 1 - h_lo_h/Weff_hi
= 1 - 2.7516/3.6799
= 25.225 %
Largest squeeze
[70] A_g_lo = b_min·h_min - 2(1 - π/4)·(g_h·r_g)²
= 4.7028·2.7516 - 0.026857
= 12.914 mm²
Smallest groove area at the hot end
[71] fill_max = (π/4)·Wh_hi²·(f_h/g_ref,h)/A_g_lo
= (π/4)·3.6747²·(1.0208/1.0006)/12.914
= 83.783 %
Largest gland fill (unstretched section, volume basis: the held ring takes its whole volume change in its section)
[72] squeeze_nom = 1 - h/W_eff at the mean dimensions
= 1 - 2.78/3.4552
= 19.541 %
Squeeze at the mean dimensions, installed
[73] sq_min_exact = exact non-linear corner (check)
= corner chosen by the sensitivities
= 15.18 %
Check: smallest squeeze at the exact corner

[74] sq_max_exact = exact non-linear corner (check)
= corner chosen by the sensitivities
= 25.176 %
Check: largest squeeze at the exact corner
[75] gap_max = largest gap over the installed, cold and hot temperatures (diametral clearance with its own expansion) plus the bore dilation under pressure
= mean 0.068 mm, worst-case maximum
= 0.10606 mm
Largest extrusion gap
[76] gap_min = smallest gap over the temperatures
= mean 0.068 mm, worst-case minimum
= 0.029984 mm
Smallest extrusion gap
[78] gap_allow = permitted extrusion gap
= entered 0.2 mm
= 0.2 mm
Permitted extrusion gap (your record or value)
[79] u_gap = gap_max / gap_allow
= 0.10606 / 0.2
= 0.53032
Largest gap against the permitted gap
[80] p_sq_min = fraction of 20000 samples with squeeze below the lower limit (cold end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: squeeze below the lower limit (cold end)
[81] cpk_sq_min = limit distance over three of the linearised model (the Monte Carlo itself model)
= = 0.0080809
= 1.5612
Capability index C_pk: squeeze below the lower limit (cold end)
[82] p_sq_max = fraction of 20000 samples with squeeze above the upper limit (hot end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: squeeze above the upper limit (hot end)
[83] cpk_sq_max = limit distance over three of the linearised model (the Monte Carlo itself model)
= = 0.0077697
= 3.4698
Capability index C_pk: squeeze above the upper limit (hot end)
[84] p_fill_max = fraction of 20000 samples with gland fill above the upper limit (hot end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: gland fill above the upper limit (hot end)
[85] cpk_fill_max = limit distance over three of the linearised model (the Monte Carlo itse model)
= = 0.01515
= 1.648
Capability index C_pk: gland fill above the upper limit (hot end)
[86] p_fill_min = fraction of 20000 samples with gland fill below the lower limit (cold end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: gland fill below the lower limit (cold end)

[87] cpk_fill_min = limit distance over three □ of the linearised model (the Monte Carlo itse
model)
= □ = 0.01376
= 2.3815
Capability index C_pk: gland fill below the lower limit (cold end)
[88] p_stretch = fraction of 20000 samples with installed stretch outside its limits
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: installed stretch outside its limits
[89] cpk_stretch = limit distance over three □ of the linearised model (the Monte Carlo itsel
model)
= □ = 0.0027575
= 3.158
Capability index C_pk: installed stretch outside its limits
[90] p_gap = fraction of 20000 samples with extrusion gap above the permitted gap
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: extrusion gap above the permitted gap
[91] cpk_gap = limit distance over three □ of the linearised model (the Monte Carlo itself ru
model)
= □ = 0.0091585
= 4.8028
Capability index C_pk: extrusion gap above the permitted gap
[92] pareto = largest contributor to the squeeze spread (worst case)
= ring cross-section W 66.373 %; ring inside diameter d 9.3264 %; bore D_s 9.6019 %; piston g
D_g 14.699 %
= "ring cross-section W"
Contribution Pareto

5. Results

SYMBOL	VALUE	DESCRIPTION
squeeze_nom	19.541 %	Squeeze at the mean dimensions
squeeze_min	15.125 %	Smallest squeeze
squeeze_max	25.225 %	Largest squeeze
fill_min	63.953 %	Smallest gland fill
fill_max	83.783 %	Largest gland fill
stretch_nom	2.6125 %	Installed stretch, mean dimensions
stretch	3.5104 %	Largest installed stretch
stretch_min	1.729 %	Smallest installed stretch
X_c	4.1575 %	Largest stretch at the cold end
u_stretch	0.58506	Largest stretch against its limit
red_nom	1.2812 %	Cross-section reduction at the mean stretch
gap_min	0.029984 mm	Smallest extrusion gap
gap_max	0.10606 mm	Largest extrusion gap (over the temperature range, with the bore dilation)
gap_allow	0.2 mm	Permitted extrusion gap

SYMBOL	VALUE	DESCRIPTION
u_gap	0.53032	Largest gap against the permitted gap
p_sq_min	0 %	Fraction outside the limit: squeeze below the lower limit (cold end)
cpk_sq_min	1.5612	C_pk: squeeze below the lower limit (cold end)
p_sq_max	0 %	Fraction outside the limit: squeeze above the upper limit (hot end)
cpk_sq_max	3.4698	C_pk: squeeze above the upper limit (hot end)
p_fill_max	0 %	Fraction outside the limit: gland fill above the upper limit (hot end)
cpk_fill_max	1.648	C_pk: gland fill above the upper limit (hot end)
p_fill_min	0 %	Fraction outside the limit: gland fill below the lower limit (cold end)
cpk_fill_min	2.3815	C_pk: gland fill below the lower limit (cold end)
p_stretch	0 %	Fraction outside the limit: installed stretch outside its limits
cpk_stretch	3.158	C_pk: installed stretch outside its limits
p_gap	0 %	Fraction outside the limit: extrusion gap above the permitted gap
cpk_gap	4.8028	C_pk: extrusion gap above the permitted gap

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Squeeze minimum within the band	squeeze_min 15.125 %	≥ 15 %	0	oring.criteria record
PASS	Squeeze maximum within the band	squeeze_max 25.225 %	≤ 30 %	0	oring.criteria record
PASS	Gland fill maximum within the band	fill_max 83.783 %	≤ 85 %	0	oring.criteria record
PASS	Gland fill minimum within the band	fill_min 63.953 %	≥ 60 %	0	oring.criteria record
PASS	Installed stretch not above the limit	stretch 3.5104 %	≤ 6 %	0	oring.criteria record; PTD 5705 installation rule
PASS	Installed stretch not above the limit at the cold end	X_c 4.1575 %	≤ 6 %	0	oring.criteria record
PASS	Installed stretch not below the limit	stretch_min 1.729 %	≥ 0 %	0	oring.criteria record
PASS	Extrusion gap within the permitted gap	gap_max 0.10606 mm	≤ 0.2 mm (entered 0.2 mm)	0	oring.extrusion record

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: alpha (Material properties); d_r, d_r_tol, W, W_tol (Ring size); q_lo_s, q_hi_s, fill_lo, fill_hi, x_lo, x_hi (Squeeze, fill and stretch bands); gap_allow (Permitted extrusion gap). Enter each as one so its source, edition and licence travel with it.
WARNING — input.face_p: face_p (Face seal only: which side the pressure acts on (external: the ring ID seats on the groove inner wall and is stretched; internal: the ring OD seats on the groove outer wall and is compressed)) was not entered; the template default 'External (ring ID seated on the groove)' is used.
WARNING — input.h_f: h_f (Face-seal groove depth (face seal only)) was not entered; the template default 2.7 mm is used.
WARNING — input.h_f_u: h_f_u (Face-seal groove depth, upper deviation) was not entered; the template default 0.05 mm is used.
WARNING — input.h_f_l: h_f_l (Face-seal groove depth, lower deviation) was not entered; the template default 0 mm is used.
WARNING — input.e_f: e_f (Face-seal extrusion gap between the flanges (face seal only)) was not entered; the template default 0.05 mm is used.
WARNING — input.e_f_u: e_f_u (Face-seal gap, upper deviation) was not entered; the template default 0.05 mm is used.
WARNING — input.e_f_l: e_f_l (Face-seal gap, lower deviation) was not entered; the template default 0 mm is used.
WARNING — input.nu: nu (Poisson ratio of the ring, for the linear relation) was not entered; the template default 0.49 is used.
WARNING — input.q_lo_d: q_lo_d (Dynamic duty: lowest acceptable squeeze) was not entered; the template default 10 % is used.
WARNING — input.q_hi_d: q_hi_d (Dynamic duty: highest acceptable squeeze) was not entered; the template default 25 % is used.
WARNING — input.od_hi: od_hi (Ring outside-diameter compression: highest acceptable, in a rod-seal groove or against the groove outer wall of an internal-pressure face seal (your record; the default is a placeholder equal to the stretch limit, not a handbook value)) was not entered; the template default 6 % is used.
WARNING — input.dD_p: dD_p (Diametral growth under the operating pressure of the bore (piston seal) or of the housing bore round the rod (rod seal), from your cylinder or housing calculation; added to the extrusion gap (0 if not known)) was not entered; the template default 0 mm is used.

8. Verification

```
Template fp.oring-gland@1.2.0 Content hash 11c67bf30e5d580d Engine calc-engine/1.0.0
Known-answer cases passed 13/13 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```